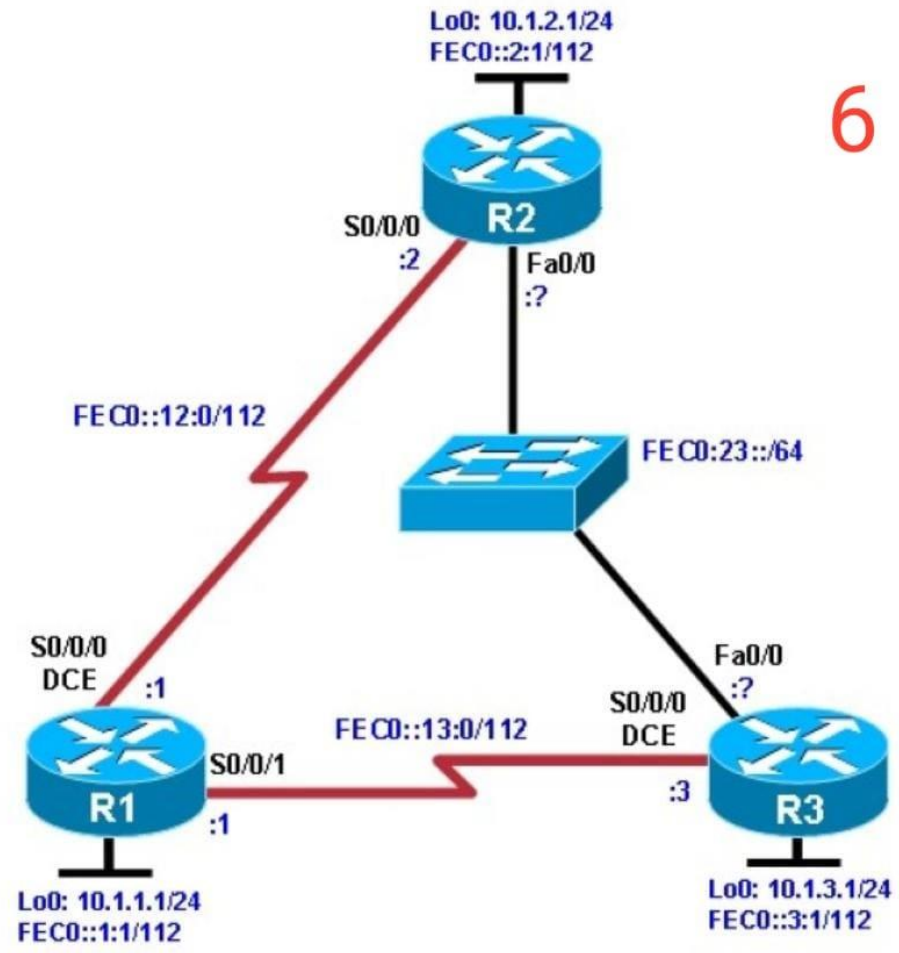
Configuring OSPF for IPv6



Step 2: Configuring the Hostname and Loopback Interfaces

● R1 Configuration

```
Router(config)# hostname R1

R1(config)# interface loopback0
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# ipv6 address FEC0::1:1/112
```

● R2 Configuration

```
Router(config)# hostname R2

R2(config)# interface loopback0
R2(config-if)# ip address 10.1.2.1 255.255.255.0
R2(config-if)# ipv6 address FEC0::2:1/112
```

● R3 Configuration

```
Router(config)# hostname R3

R3(config)# interface loopback0
R3(config-if)# ip address 10.1.3.1 255.255.255.0
R3(config-if)# ipv6 address FEC0::3:1/112
```

Step 3: Configure Static IPv6 Addresses

● R1

```
R1(config)# interface serial0/0/0
R1(config-if)# ipv6 address FEC0::12:1/112
R1(config-if)# clockrate 64000
R1(config-if)# bandwidth 64
R1(config-if)# no shutdown

R1(config)# interface serial0/0/1
R1(config-if)# ipv6 address FEC0::13:1/112
R1(config-if)# bandwidth 64
R1(config-if)# no shutdown
```

● R2

```
R2(config)# interface serial0/0/0
R2(config-if)# ipv6 address FEC0::12:2/112
R2(config-if)# bandwidth 64
R2(config-if)# no shutdown
```

● R3

```
R3(config)# interface serial0/0/0
R3(config-if)# ipv6 address FEC0::13:3/112
R3(config-if)# clockrate 64000
R3(config-if)# bandwidth 64
R3(config-if)# no shutdown
```

Connectivity Testing

```
R1# ping FEC0::12:2
R1# ping FEC0::13:3

R2# ping FEC0::12:1
```

```
R3# ping FEC0::13:1
```

Step 4: Change the Link-Local Address on an Interface

Check Current Interface

```
R1# show ipv6 interface serial 0/0/0
R2# show ipv6 interface serial 0/0/0
```

Configure Link-Local Address

● R1

```
R1(config)# interface serial0/0/0
R1(config-if)# ipv6 address FE80::1 link-local
```

● R2

```
R2(config)# interface serial0/0/0
R2(config-if)# ipv6 address FE80::2 link-local
```

Test Connectivity

```
R1# ping FE80::2
R2# ping FE80::1
```

Verify

```
R1# show ipv6 interface serial 0/0/0
R2# show ipv6 interface serial 0/0/0
```

Step 5: Configure EUI-64 Addresses

● R2

```
R2(config)# interface fastEthernet 0/0
R2(config-if)# ipv6 address FEC0:23::/64 eui-64
R2(config-if)# no shutdown
```

● R3

```
R3(config)# interface fastEthernet 0/0
R3(config-if)# ipv6 address FEC0:23::/64 eui-64
R3(config-if)# no shutdown
```

Verification

```
R2# show ipv6 interface fastEthernet 0/0
R2# show ipv6 interface brief
```

Connectivity Test

```
R2# ping FEC0:23::218:B9FF:FECD:BEF0
R3# ping FEC0:23::218:B9FF:FE92:28D8
```

Step 6: Enable IPv6 Routing and CEF

● R1

```
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 cef
```

● R2

```
R2(config)# ipv6 unicast-routing
R2(config)# ipv6 cef
```

● R3

```
R3(config)# ipv6 unicast-routing
R3(config)# ipv6 cef
```

Step 7: Configure OSPFv3

● R1

```
R1(config)# interface loopback0
R1(config-if)# ipv6 ospf 1 area 0

R1(config)# interface serial0/0/0
R1(config-if)# ipv6 ospf 1 area 0

R1(config)# interface serial0/0/1
R1(config-if)# ipv6 ospf 1 area 0
```

● R2

```
R2(config)# interface loopback0
R2(config-if)# ipv6 ospf 1 area 0

R2(config)# interface serial0/0/0
R2(config-if)# ipv6 ospf 1 area 0

R2(config)# interface fastEthernet 0/0
R2(config-if)# ipv6 ospf 1 area 0
```

● R3

```
R3(config)# interface loopback0
R3(config-if)# ipv6 ospf 1 area 0

R3(config)# interface serial0/0/0
R3(config-if)# ipv6 ospf 1 area 0

R3(config)# interface fastEthernet 0/0
R3(config-if)# ipv6 ospf 1 area 0
```

● Verification Commands

Run on any router:

```
show ipv6 ospf neighbor
show ipv6 route
```

Example:

```
R1# show ipv6 ospf neighbor
R2# show ipv6 ospf neighbor
R3# show ipv6 ospf neighbor

R1# show ipv6 route
R2# show ipv6 route
R3# show ipv6 route
```

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R1 Router

■ Copy-paste this entire block in R1

```
enable
configure terminal
```

```
hostname R1

interface loopback0
ip address 10.1.1.1 255.255.255.0
ipv6 address FEC0::1:1/112
exit

interface serial0/0/0
ipv6 address FEC0::12:1/112
clock rate 64000
bandwidth 64
no shutdown
exit

interface serial0/0/1
ipv6 address FEC0::13:1/112
bandwidth 64
no shutdown
exit

interface serial0/0/0
ipv6 address FE80::1 link-local
exit

ipv6 unicast-routing
ipv6 cef

interface loopback0
ipv6 ospf 1 area 0
exit

interface serial0/0/0
ipv6 ospf 1 area 0
exit

interface serial0/0/1
ipv6 ospf 1 area 0
exit

end
```

R2 Router

■ Copy-paste this entire block in R2

```
enable
configure terminal
hostname R2

interface loopback0
ip address 10.1.2.1 255.255.255.0
ipv6 address FEC0::2:1/112
exit

interface serial0/0/0
ipv6 address FEC0::12:2/112
bandwidth 64
```

```
no shutdown
exit

interface serial0/0/0
ipv6 address FE80::2 link-local
exit

interface fastethernet0/0
ipv6 address FEC0:23::/64 eui-64
no shutdown
exit

ipv6 unicast-routing
ipv6 cef

interface loopback0
ipv6 ospf 1 area 0
exit

interface serial0/0/0
ipv6 ospf 1 area 0
exit

interface fastethernet0/0
ipv6 ospf 1 area 0
exit

end
```

R3 Router

■ Copy-paste this entire block in R3

```
enable
configure terminal
hostname R3

interface loopback0
ip address 10.1.3.1 255.255.255.0
ipv6 address FEC0::3:1/112
exit

interface serial0/0/0
ipv6 address FEC0::13:3/112
clock rate 64000
bandwidth 64
no shutdown
exit

interface fastethernet0/0
ipv6 address FEC0:23::/64 eui-64
no shutdown
exit

ipv6 unicast-routing
ipv6 cef
```

```
interface loopback0
ipv6 ospf 1 area 0
exit

interface serial0/0/0
ipv6 ospf 1 area 0
exit

interface fastethernet0/0
ipv6 ospf 1 area 0
exit

end
```

Verification Commands (Run when required)

You **copy these only when verification is asked**:

```
ping FEC0::12:2
ping FEC0::13:3
ping FE80::2
ping FE80::1

show ipv6 interface brief
show ipv6 interface serial 0/0/0
show ipv6 ospf neighbor
show ipv6 route
```